Knitted HexOct: is the c/h stacking a design dial?

Lattice mechanics and a see-through/airflow proxy for c, h and mixed stacking sequences



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In the Knitted HexOct, which DICTO proposed, each knitted sheet is one close-packed layer of nodes. Links join every node to the 3 nearest nodes of the next sheet. At a c sheet the links run straight on; at an h sheet they fold back in a chevron. All link lengths, angles and node counts are the same whatever the stacking, so all stackings have the same density. That makes the stacking sequence a candidate *design dial*: the same yarn and the same stitch count, but a different property, depending only on where the links turn. This study checks that idea in idealised lattice models.

> **Result.** Through-thickness compression, buckling onset and rigidity are the same in every sequence (to 4 digits). On those properties c/h is no dial. Two properties do change strongly: (1) the **straight see-through area** along the stacking axis is 34.7 % in 2H, 9.9 % in 4H and 3.25 % in 3C at link radius 0.05 a, with equal porosity (10.7×). At radius 0.1 a it is 14.3 % in 2H and 0 in 3C. (2) **Free transverse shear stiffness** changes when the sheets are soft in-plane: 3C/2H = 0.057 (17.5×) with links only and rigid joints. Graded stacks fall in between. The ratio rises to 0.900 when the sheets are as stiff as the links. The controlling number for shear is the stack's net c-twist (mean Hägg symbol), not its share of c sheets.

1 Model

Geometry (from the spec, studies/2026-11-02-knitted-hexoct): node spacing a = 1, sheet spacing √(2/3), links of length 1 at 54.7356°. Each block has 12 sheets with 6 × 6 nodes per sheet and is periodic in-plane. A program checks every node: 3 links up, 3 links down and 6 in-sheet neighbours. The sequences are written as c/h labels of the 10 inner sheets: 3C (all c), 2H (all h), 4H (hc…), 6H (hcc…), Fibonacci (abaababaab, a → c, b → h), and two graded stacks (5 h at the bottom and 5 c at the top, and the reverse). **Members** are round rods, radius r = 0.05 a, E_s = 1, ν = 0.3. They are modelled as pin-jointed bars and as rigid-jointed 3D Euler–Bernoulli frame members. There are three sheet variants: *links only* (the sheets are just nodes); *stiff sheets* (in-sheet triangular members 100× stiffer than the links, our stand-in for nearly rigid sheets); and *full* (in-sheet members the same as the links; 3C is then the octet truss, 2H the hcp truss). **Tests on the block:** compression with bonded platens (C33) and with lubricated platens and free lateral strain (E3); transverse shear with bonded platens and with free in-plane strain (the bulk limit is $1/S_{55}$); and in-plane E1, E2 and G12 of the free plate. The 3D-periodic 6×6 stiffness matrix C is computed for 3C, 2H, 4H and 6H. **Check:** for 3C (full, pin joints) the computed E[100] equals $\bar{\rho}/9$, the octet-truss value of Deshpande, Fleck & Ashby (2001). The computed ratio is 1.0000000000. **Energy proxy:** an event-to-event analysis with pin joints. Each strut buckles at the pinned Euler load $\pi^2 E_s I/a^2$ and then carries that load unchanged; tension stays elastic. We report the work done up to 5 % strain. **Airflow proxy:** rods are capsules (round ends). The straight see-through fraction is exact: a line along the stacking axis misses a capsule exactly when its plan point lies more than r from the capsule's plan segment. Tortuosity is the shortest open path from each open point of the top plane to the bottom plane, divided by the thickness. It is computed on a 26-neighbour voxel graph, h = 0.02 a, checked at 0.0125 a.

2 Properties per sequence

sequence	word	(Hägg)	C33 bonded	E3 free	G13 bonded	G13 free links only	G13 free full	E1 in-plane links only	work 5 % shear xz	see-through r = 0.05	see-through r = 0.1	tortuosity r = 0.05 / 0.1
3C	ABCABCABCABC	1.000	14.865	0.492	3.715	0.193	3.360	0.051	3.50	3.25 %	0	1.003 / 1.039
2H	ABABABABABAB	0.091	14.865	0.492	3.715	3.378	3.699	0.063	3.28	34.7 %	14.3 %	1.001 / 1.008
4H	ABACABACABAC	−0.091	14.865	0.492	3.715	3.283	3.699	0.055	3.28	9.9 %	0	1.002 / 1.021
6H	ABACBCABACBC	−0.091	14.865	0.492	3.715	3.276	3.699	0.055	3.28	3.25 %	0	1.003 / 1.033
Fibonacci	ABCBACABACBC	−0.091	14.865	0.492	3.715	3.275	3.699	0.055	3.28	3.25 %	0	1.003 / 1.032
graded c top / h bottom	ABABABACBACB	−0.455	14.865	0.492	3.715	0.927	3.626	0.057	3.28	3.25 %	0	1.003 / 1.022
graded h top / c bottom	ABCABCACACAC	0.455	14.865	0.492	3.715	0.927	3.626	0.057	3.28	3.25 %	0	1.003 / 1.027

Stiffness in 10^{-3} E_s. C33, E3 and G13 bonded: rigid joints, links only (relative density $\bar{\rho}$ = 0.0305). These values are the same for all sequences in every variant; with in-sheet members they are 14.810–14.865, 13.44–13.52 and 3.70–3.76 ($\bar{\rho}$ = 0.0639). "Full": pin joints, links + in-sheet members. Work done in xz shear up to 5 % strain, in 10^{-6} E_s, full, pin joints. In compression the first buckling happens at strain 0.0093 (bonded) and 0.0102 (lubricated), at stress 1.370×10^{-4} E_s, in every sequence; there all 1188 links buckle together. Porosity 0.9406 (r = 0.05) and 0.7869 (r = 0.1) in every sequence. Full table: data/tables.txt.

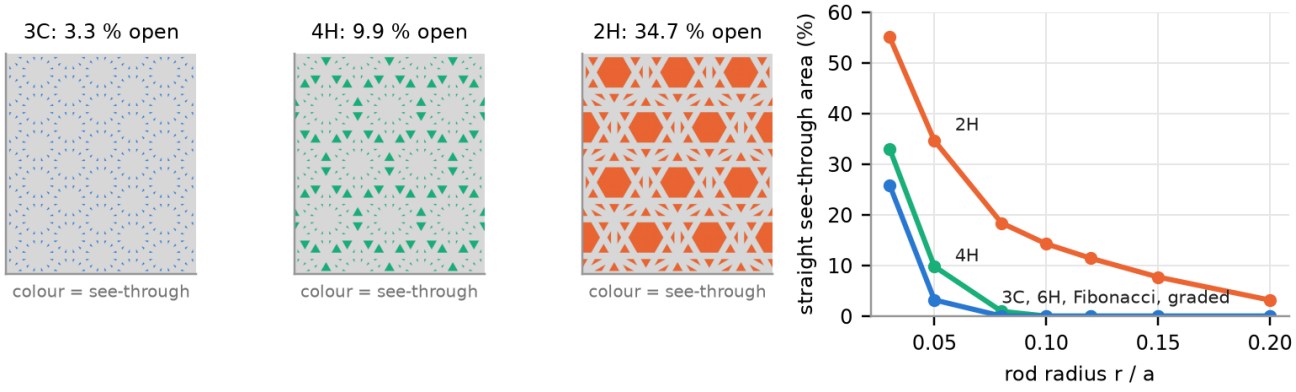


Figure 1. Straight see-through area along the stacking axis. Left: open plan area (colour) over 3 × 2 rectangular cells at link radius 0.05 *a*. Right: open area against rod radius. The open area depends only on which of the three link plan pairs AB, BC, CA occur: 2H has one, 4H two, and 3C, 6H, Fibonacci and the graded stacks have all three.

3 Rigidity: what h does

Links only, pin joints. Every node has 6 bars, so the periodic cell has exactly 3N bars, the Maxwell isostatic count, in every sequence. The compatibility matrices show the same picture in every sequence: 3 macroscopic strains cost no bar stretching. These are the in-plane strains combined with $\varepsilon_{zz} = -(\varepsilon_{xx}+\varepsilon_{yy})/2$; in 3C they are mixed with transverse shear. The 6×6×12 periodic cell also has periodic mechanisms: 33 (3C), 37 (4H), 39 (2H), 51 (6H). Each comes with exactly as many self-stress states plus 3, as Guest & Hutchinson (2003) show must happen in an isostatic repetitive frame. The finite 12-sheet block has 105 mechanisms in every sequence. So the link net alone is not rigid, whether c or h. It carries through-thickness compression only between bonded faces. **With in-sheet members** every sequence is rigid (0 mechanisms, 1191 self-stress states in the block). The hcp-type h stack is as rigid and stretch-dominated as the octet-type c stack: C33, C13 and C44 are equal, and E3 is identical.

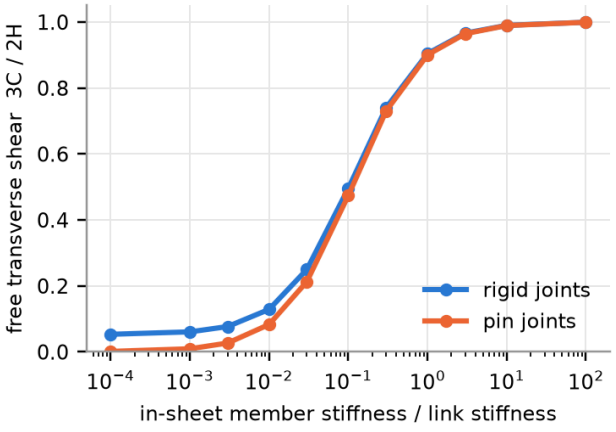


Figure 2. Shear dial strength (periodic bulk, $1/S_{55}$ of 3C over 2H) against the stiffness of the in-sheet members relative to the links. Soft sheets give a strong dial (0.05 rigid joints, 0.001 pin joints at 10^{-4}). Sheets as stiff as the links give 0.90, and 100× stiffer sheets 0.999.

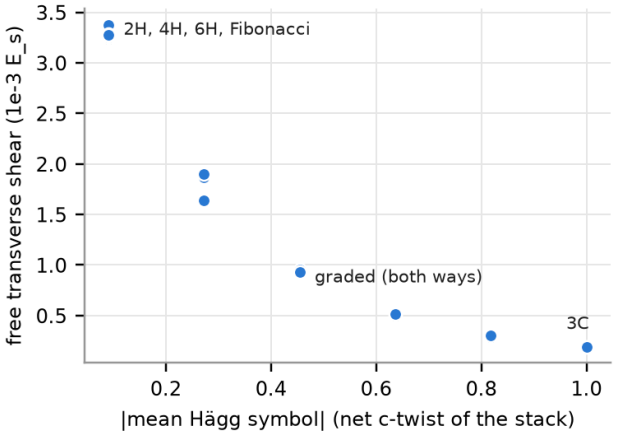


Figure 3. Free transverse shear stiffness (links only, rigid joints, 12-sheet block) against |mean Hägg symbol| for the 7 sequences, 11 bottom-c/top-h stacks and 5 stacks of c runs split by single h sheets. An h sheet reverses the twist sense, so c runs split by an h cancel: ccccchcccc gives 3.24, close to 2H.

4 Why shear changes and compression does not

The c stack has 3-fold (trigonal) symmetry about the stacking axis. Its stiffness matrix therefore has a shear-stretch term $C_{14} = 1.31\times10^{-3}\ E_s$: transverse shear is coupled to in-plane stretching and in-plane shear. In 2H, 4H and 6H, $C_{14} = 0$, as hexagonal symmetry requires. When the faces are free to strain in-plane, the 3C fabric can give way in shear by straining its sheets. If the sheets are soft, that costs little, so $1/S_{55}$ drops well below C_{55} (roughly by coupling² / in-plane sheet stiffness). With bonded platens the in-plane strain is blocked and every sequence gives the same value. Through-thickness compression loads all links equally (each at $\cos^2 = 2/3$ of the strain) whether they run straight on or fold back. So C33, E3 and the buckling onset do not depend on the stacking.

5 Conclusion and possible uses

Yes, c/h is a usable dial, for two properties only. It sets the straight see-through area: 10.7× from 3C to 2H at $r = 0.05\ a$, at equal porosity, mass and thickness. It also sets the free transverse shear stiffness, by up to 17.5× when the sheets are soft and laterally free. It does **not** set compression stiffness, buckling onset, compression energy or pin-jointed rigidity. Possible uses, with what the model does and does not support:

- **Insulation and breathable layers:** a c-rich stack closes the straight sight-lines through the thickness, and a 2H stack keeps one third of the area open. This is a line-of-sight result, relevant to radiation and light passing straight through. We did not compute heat transfer or permeability. Tortuosity barely changes (1.001–1.039).
- **Padding and footwear:** cushioning stiffness in compression is not changed by c/h in this model. Shear compliance (sole and heel shear, comfort layers) is changed, if the knitted sheets are much softer in-plane than the links, as knits usually are. 3C couples shear to in-plane strain; graded stacks give intermediate shear stiffness through the thickness.
- **Composite preforms:** once resin stiffens the sheets, the shear dial falls to about 10 % (ratio 0.900 at equal stiffness). The open straight z-channels of 2H could matter for through-thickness infiltration, but no flow solve was made.

Model limits. Idealised geometry: straight rods meeting at points, every link exactly length a. No yarn mechanics: no loops, compaction, friction, yarn twist, or yarns that cannot carry compression. Linear elasticity, apart from the buckling plateau of the energy proxy; no post-buckling softening; no contact. A finite block of 12 sheets, periodic in-plane. Results are given relative to the rod modulus E_s and depend on r/a. Airflow is a geometric proxy (straight see-through area and shortest-path tortuosity), not a permeability. Physical tests of knitted samples (T-c and T-h of the spec) are needed before any application claim.

6 Prior art and sources checked (2 November 2026)

Octet-truss stiffness and the stretch-dominated/bending-dominated distinction are from Deshpande, Fleck & Ashby [1, 2]; our 3C check reproduces [1]. The mechanism and self-stress counts follow Guest & Hutchinson [3]. Spacer fabrics with one gap are established cushioning materials [4]. 3D woven, braided, stitched and knitted preforms are reviewed in [5, 6]. Lattices that copy crystal grains and twins ("meta-grains") were made by Pham et al. [7]. In atomic polytypes the elastic constants vary almost linearly with hexagonality [8]. In the results we saw, we found no mechanical comparison of hcp-type with fcc-type (octet) truss lattices, and none of mixed c/h stackings as a design variable. This is not a full literature review.

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2. V. S. Deshpande, M. F. Ashby, N. A. Fleck, Foam topology: bending versus stretching dominated architectures, Acta Mater. 49 (2001) 1035–1040. Known reference, not opened.
3. S. D. Guest, J. W. Hutchinson, On the determinacy of repetitive structures, J. Mech. Phys. Solids 51 (2003) 383–391. Record seen.
4. Y. Liu, H. Hu, L. Zhao, H. Long, Compression behavior of warp-knitted spacer fabrics for cushioning applications, Textile Res. J. 82 (2012) 11–20; also Y. Liu, H. Hu, TRJ 86 (2016) 3–12. Abstracts and records seen.
5. A. P. Mouritz et al., Review of applications for advanced three-dimensional fibre textile composites (1999). Abstract seen; journal record not confirmed in the search.
6. L. Tong, A. P. Mouritz, M. K. Bannister, 3D Fibre Reinforced Polymer Composites, Elsevier 2002. Book listing seen.
7. M.-S. Pham, C. Liu, I. Todd, J. Lertthanasarn, Damage-tolerant architected materials inspired by crystal microstructure, Nature 565 (2019) 305–311. Records seen.
8. M. Keller et al., First-principles insight in structure-property relationships of hexagonal Si and Ge polytypes, arXiv 2302.10822 (2023). Library copy, pp. 1 and 11 read.

Also checked, not cited: library: Lalvani 1990, NASA TM-102635 (layered oct-tet truss part counts; pp. 1–3 read; no mechanics); Goyal & Lakes 2026 and Korshunova et al. 2020/2021 (octet truss; catalogue entries only); Krishnamurthy, Akleman et al. 2020/2022 (woven tiles; snippets only); Rodriguez-Horta et al. 2017 (close-packed layer interactions; catalogue entries only). Library queries: octet truss stretch-dominated; spacer fabric compression; 3D woven/knitted preform; hcp/fcc stacking elastic constants polytypes; Maxwell isostatic periodic lattice mechanisms. Web queries: the Deshpande–Fleck–Ashby octet paper; Guest–Hutchinson determinacy; warp-knitted spacer fabric compression; the Mouritz 1999 review; "hcp truss lattice vs fcc octet truss lattice stiffness stacking sequence"; the Pham 2019 meta-crystals paper. Scripts, data and full tables: study folder 2026-11-02-hexoct-knit-mechanics (run on dicto-big-brother).